

MOST REPEATED MATHEMATICS QUESTIONS

Fixed Patterns & High-Weightage Templates for Section D & E

Created for Thunderstudy By Aashi Mathematics

Based on the analysis of the provided question papers, master these templates to cover a massive portion of the paper.

1. Matrices: The "System of Linear Equations" Question (5 Marks)

- **The Standard Question:** You are given a system of three linear equations (e.g., $x - 2y = 10$, $2x - y - z = 8$) and asked to solve them using matrices.
- **The "Product" Variation:** You are given two matrices, A and B. You are asked to find the product AB, show that it equals a scalar matrix (like $8I$), use that to find the inverse, and then solve a system of equations.
- **The Word Problem Variation:** A scenario involves production numbers (e.g., "furniture workshop producing chairs, tables, and beds"). You must first convert the text into linear equations and then solve using matrices.

2. 3D Geometry: The "Shortest Distance" Question (5 Marks)

- **Shortest Distance:** You are given equations of two lines in vector form ($r = a_1 + \lambda b_1$ and $r = a_2 + \mu b_2$) and asked to find the shortest distance between them.
- **Foot of Perpendicular / Image:** A variation asks you to find the coordinates of the "foot of the perpendicular" drawn from a point to a line, or the "image" of that point in the line.

3. Integrals: Area Under the Curve (5 Marks)

- **Specific Shapes:** Find the area bounded by an **ellipse** ($x^2/a^2 + y^2/b^2 = 1$) or a **parabola** bounded by a line ($x^2 = 4y$ and $x = 4y - 2$).

- **Modulus Functions:** Find the area bounded by absolute value functions, such as $|x| + |y| = 1$ or curves involving $|x-1|$.

4. Linear Programming: The "Maximize/Minimize Z" Question (5 Marks)

- **Graphical Solution:** You are given an objective function $Z = ax + by$ and a set of constraints ($x + y \leq 10$, etc.). You must graph the feasible region, find the corner points, and determine the maximum or minimum value.

5. Probability: Bayes' Theorem (Case Study / 4 Marks)

- **The Scenario:** A factory has two machines producing items with different defect rates, or a population is tested for a disease with a specific accuracy rate.
- **The Question:** "Given that the result is positive/defective, what is the probability it came from Machine A / Person has the disease?" This requires **Bayes' Theorem**.

6. Differential Equations (3 or 5 Marks)

- **Linear Differential Equation:** Solve the form $dy/dx + Py = Q$. You must find the Integrating Factor (I.F. = $e^{\int P dx}$).
- **Homogeneous Equation:** Solve equations where y/x substitution is required, often given as $(x^2 + y^2)dx - 2xy dy = 0$.

7. Relations and Functions (Case Study / 4 Marks)

- **Equivalence Check:** You are given a real-life relation (e.g., "students in a library," "lines on a track"). You must check if the relation is **Reflexive, Symmetric, and Transitive**.
- **One-One / Onto:** A standard question asks to prove if a function is Injective (One-One) or Surjective (Onto).

Most Repeated Questions By Aashi Mathematics
Analyze. Practice. Conquer.